

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial}{\partial \phi} \right) + \frac{1}{r^2 \sin^2 \phi} \frac{\partial^2}{\partial \theta^2} \quad (1.24)$$

are the material derivative and Laplacian operators in spherical coordinates.

1.3 FIRST LAW OF THERMODYNAMICS

The preceding two principles—mass conservation and force balance—are in many cases sufficient for solving the *flow part* of the convective heat transfer

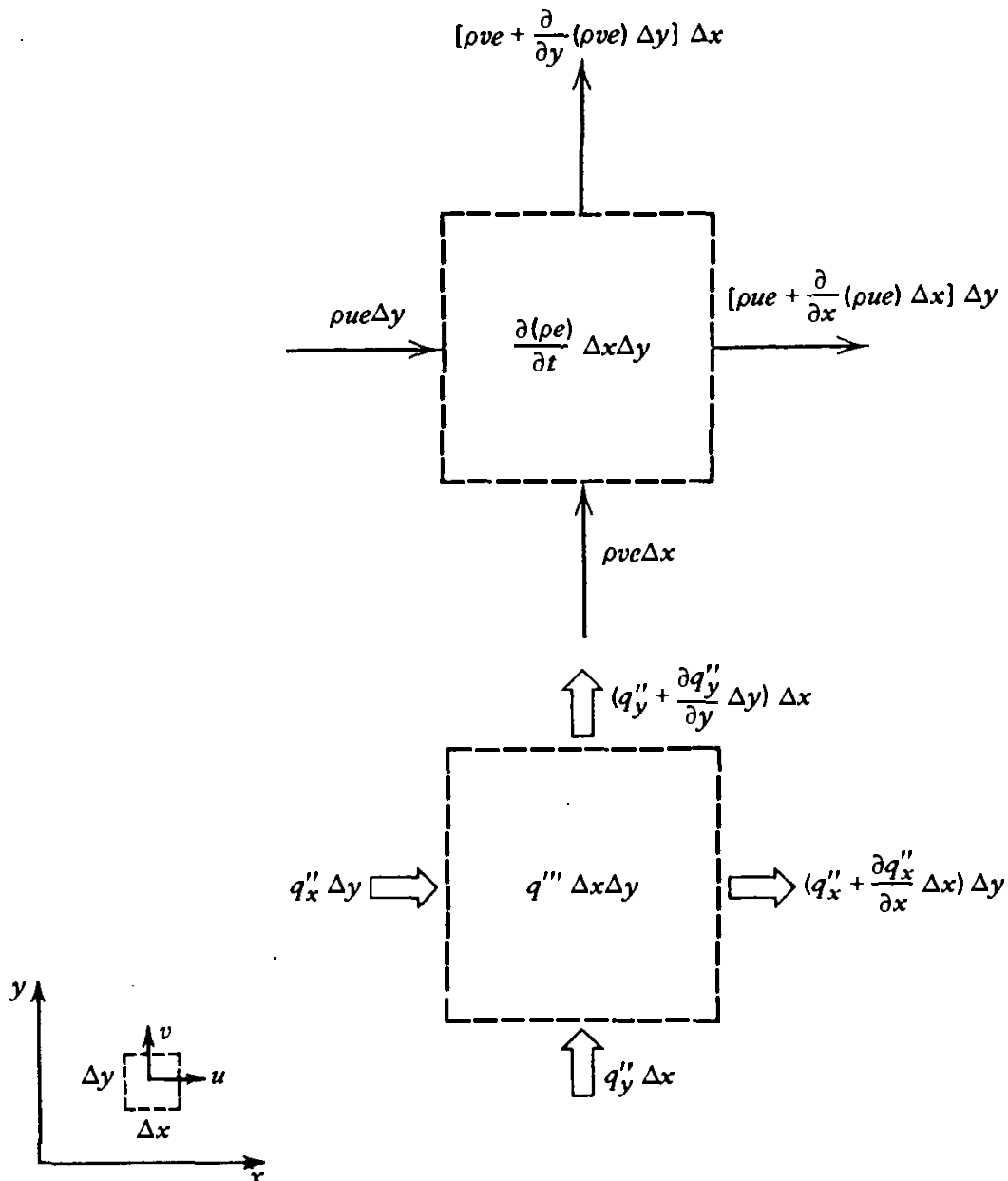


Figure 1.3 First law of thermodynamics applied to a control volume in two-dimensional flow (for work transfer, see Fig. 1.2).

problem; note at this juncture the availability of four equations (mass conservation plus three force balances) for determining four unknowns (three velocity components plus pressure). The exception to this statement is the subject of Chapter 4, where the natural flow is driven by the heat administered to the flowing fluid. In all cases, however, the *heat transfer part* of the convection problem requires a solution for the temperature distribution through the flow, especially in the close vicinity of the solid walls bathed by the heat-carrying fluid stream (Chapter 2). The additional equation for accomplishing this ultimate objective is the first law of thermodynamics or the *energy equation*.

For the control volume of finite size $\Delta x \Delta y$ in Fig. 1.3, the first law of thermodynamics requires that

$$\begin{aligned}
 & \left(\begin{array}{l} \text{rate of energy} \\ \text{accumulation in the} \\ \text{control volume} \end{array} \right)_1 \\
 &= \left(\begin{array}{l} \text{net transfer of} \\ \text{energy by fluid flow} \end{array} \right)_2 + \left(\begin{array}{l} \text{net heat transfer} \\ \text{by conduction} \end{array} \right)_3 \\
 &+ \left(\begin{array}{l} \text{rate of internal} \\ \text{heat generation (e.g.,} \\ \text{electrical power} \\ \text{dissipation)} \end{array} \right)_4 - \left(\begin{array}{l} \text{net work transfer} \\ \text{from the control} \\ \text{volume to its} \\ \text{environment} \end{array} \right)_5 \quad (1.25)
 \end{aligned}$$

According to the energy flow diagrams sketched in Fig. 1.3, the groups of terms above are

$$\begin{aligned}
 \{\cdot\}_1 &= \Delta x \Delta y \frac{\partial}{\partial t} (\rho e) \\
 \{\cdot\}_2 &= -(\Delta x \Delta y) \left[\frac{\partial}{\partial x} (\rho u e) + \frac{\partial}{\partial y} (\rho v e) \right] \\
 \{\cdot\}_3 &= -(\Delta x \Delta y) \left(\frac{\partial q_x''}{\partial x} + \frac{\partial q_y''}{\partial y} \right) \\
 \{\cdot\}_4 &= (\Delta x \Delta y) q''' \\
 \{\cdot\}_5 &= (\Delta x \Delta y) \left(\sigma_x \frac{\partial u}{\partial x} - \tau_{xy} \frac{\partial u}{\partial y} + \sigma_y \frac{\partial v}{\partial y} - \tau_{yx} \frac{\partial v}{\partial x} \right) \\
 &+ (\Delta x \Delta y) \left(u \frac{\partial \sigma_x}{\partial x} - u \frac{\partial \tau_{xy}}{\partial y} + v \frac{\partial \sigma_y}{\partial y} - v \frac{\partial \tau_{yx}}{\partial x} \right) * \quad (1.25')
 \end{aligned}$$

where e , q''_x , q''_y , and q''' are the specific internal energy, heat flux in the x direction, heat flux in the y direction, and dissipation rate or rate of internal heat generation.

The origin of the dissipation rate term $\{\cdot\}_5$ lies in the work transfer effected by the normal and tangential stresses sketched in Fig. 1.2b. For example, the work done per unit time by the normal stresses σ_x on the left side of the $\Delta x \Delta y$ element is negative and equal to the force acting on the boundary ($\sigma_x \Delta y$) times the boundary displacement per unit time (u), which yields $-u\sigma_x \Delta y$. Similarly, the work transfer rate associated with normal stresses acting on the right side of the element is positive and equal to $[\sigma_x + (\partial\sigma_x/\partial x) \Delta x][u + (\partial u/\partial x) \Delta x] \Delta y$. The net work transfer rate due to these two contributions is $[\sigma_x(\partial u/\partial x) + u(\partial\sigma_x/\partial x)](\Delta x \Delta y)$, as shown in the $\{\cdot\}_5$ term of eq. (1.25').

Three more work transfer rates can be calculated in the same manner by examining the effect of the remaining three stresses, τ_{xy} in the x direction and σ_y and τ_{yx} in the y direction. In the $\{\cdot\}_5$ expression above, the eight terms have been separated into two groups. It can be shown that the group denoted as $(\cdot)_*$ reduces to $-\rho(D/Dt)(u^2 + v^2)/2$, which represents the change in kinetic energy of the fluid packet; in the present treatment, this change is considered negligible relative to the internal energy change $\partial(\rho e)/\partial t$ appearing in $\{\cdot\}_1$.

Assembling expressions (1.25) into the energy conservation statement that preceded them, and using constitutive relations (1.15) and (1.16), we obtain

$$\rho \frac{De}{Dt} + e \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) = -\nabla \cdot \mathbf{q}'' + q''' - P \nabla \cdot \mathbf{v} + \mu \Phi \quad (1.26)$$

where $\mathbf{q}''$ is the heat flux vector (q''_x, q''_y) and Φ is the *viscous dissipation function*, shown later in eq. (1.45a). The quantity between parentheses on the left-hand side of eq. (1.26) is equal to zero [cf. eq. (1.6)]. In the special case where the flow can be modeled as incompressible and two-dimensional, the viscous dissipation function reduces to

$$\Phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 \quad (1.27)$$

To express eq. (1.26) in terms of enthalpy, we use the thermodynamics definition $h = e + (1/\rho)P$; hence,

$$\frac{Dh}{Dt} = \frac{De}{Dt} + \frac{1}{\rho} \frac{DP}{Dt} - \frac{P}{\rho^2} \frac{D\rho}{Dt} \quad (1.28)$$

In addition, we can express the directional heat fluxes q''_x and q''_y in terms of the local temperature gradients; that is, we invoke the *Fourier law of heat conduction*.

$$\mathbf{q}'' = -k \nabla T \quad (1.29)$$

Then, combining eqs. (1.26), (1.28), and (1.29) in the manner desired, we obtain

$$\rho \frac{Dh}{Dt} = \nabla \cdot (k \nabla T) + q''' + \frac{DP}{Dt} + \mu \Phi - \frac{P}{\rho} \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) \quad (1.30)$$

Finally, we learn from the mass conservation equation (1.6) that the last terms in parentheses in eq. (1.30) add up to zero; in conclusion, the first law of thermodynamics reduces to

$$\rho \frac{Dh}{Dt} = \nabla \cdot (k \nabla T) + q''' + \frac{DP}{Dt} + \mu \Phi \quad (1.31)$$

In order to express the energy equation (1.31) in terms of temperature, it is tempting to replace the specific enthalpy on the left-hand side by the product of specific heat $\times$ temperature. This move is correct only in cases where the fluid behaves like an ideal gas (see the ideal gas model, Table 1.1). In general, the change in specific enthalpy for a single-phase substance is expressed by the canonical relation for enthalpy [3],

$$dh = T ds + \frac{1}{\rho} dP \quad (1.32)$$

where T is the absolute temperature and ds the specific entropy change,

$$ds = \left(\frac{\partial s}{\partial T} \right)_P dT + \left(\frac{\partial s}{\partial P} \right)_T dP \quad (1.33)$$

From the last of Maxwell's relations [3, p. 173], we have

$$\left(\frac{\partial s}{\partial P} \right)_T = - \left[\frac{\partial(1/\rho)}{\partial T} \right]_P = \frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_P = -\frac{\beta}{\rho} \quad (1.34)$$

where β is the coefficient of thermal expansion,

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P \quad (1.35)$$

Table 1.1 also shows that

Table 1.1 Summary of thermodynamic relations^a and models

	Internal Energy $du = T ds - P dv$	Enthalpy $dh = T ds + v dP$	Entropy $ds = \frac{1}{T} du + \frac{P}{T} dv$
Pure substance	$du = c_v dT + \left[T \left(\frac{\partial P}{\partial T} \right)_v - P \right] dv$	$dh = c_p dT + \left[-T \left(\frac{\partial v}{\partial T} \right)_P + v \right] dP$	$ds = \frac{c_p}{T} dT - \left(\frac{\partial v}{\partial T} \right)_P dP$ $= \frac{c_v}{T} dT + \left(\frac{\partial P}{\partial T} \right)_v dv$
Ideal gas	$du = c_v dT$	$dh = c_p dT$	$ds = c_p \frac{dT}{T} - R \frac{dP}{P}$ $= c_v \frac{dT}{T} + R \frac{dv}{v}$ $= c_v \frac{dP}{P} + c_p \frac{dv}{v}$ $ds = c \frac{dT}{T}$
Incompressible liquid	$du = c dT$	$dh = c dT + v dP$	

Source: Ref. 3.

^aAccording to the classical thermodynamics notation, v is the specific volume, $v = 1/\rho$, and u is the internal energy (e in the text).

$$\left(\frac{\partial s}{\partial T}\right)_P = \frac{c_P}{T} \quad (1.36)$$

Together, eqs. (1.32)–(1.36) state that

$$dh = c_P dT + \frac{1}{\rho} (1 - \beta T) dP \quad (1.37)$$

in other words, the left-hand side of the energy equation (1.31) is

$$\rho \frac{Dh}{Dt} = \rho c_P \frac{DT}{Dt} + (1 - \beta T) \frac{DP}{Dt} \quad (1.38)$$

The “temperature” formulation of the first law of thermodynamics is therefore

$$\rho c_P \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + q''' + \beta T \frac{DP}{Dt} + \mu \Phi \quad (1.39)$$

with the following special forms:

Ideal gas ($\beta = 1/T$):

$$\rho c_P \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + q''' + \frac{DP}{Dt} + \mu \Phi \quad (1.40)$$

Incompressible liquid ($\beta = 0$):

$$\rho c \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + q''' + \mu \Phi \quad (1.41)$$

Most of the convection problems addressed in this book obey an even simpler model: namely, constant fluid conductivity k , zero internal heat generation q''' , negligible viscous dissipation $\mu \Phi$, and negligible compressibility effect $\beta T DP/Dt$. The energy equation for this model is

$$\rho c_P \frac{DT}{Dt} = k \nabla^2 T \quad (1.42)$$

or, in terms of specific coordinate systems (Fig. 1.1):

Cartesian (x,y,z):

$$\rho c_P \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (1.43a)$$

Cylindrical (r,θ,z):

$$\begin{aligned} \rho c_P \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) \\ = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] \end{aligned} \quad (1.43b)$$

Spherical (r,φ,θ):

$$\begin{aligned} \rho c_P \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\phi}{r} \frac{\partial T}{\partial \phi} + \frac{v_\theta}{r \sin \phi} \frac{\partial T}{\partial \theta} \right) \\ = k \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial T}{\partial \phi} \right) + \frac{1}{r^2 \sin^2 \phi} \frac{\partial^2 T}{\partial \theta^2} \right] \end{aligned} \quad (1.43c)$$

If the fluid can be modeled thermodynamically as an incompressible liquid, then, as in eq. (1.41), the specific heat at constant pressure c_P is replaced by the lone specific heat of the incompressible liquid, c (Table 1.1).

When dealing with extremely viscous flows of the type encountered in lubrication problems or the piping of crude oil, the model above is improved by taking into account the internal heating due to viscous dissipation,

$$\rho c_P \frac{DT}{Dt} = k \nabla^2 T + \mu \Phi \quad (1.44)$$

In three dimensions, the general viscous dissipation function can be expressed as follows:

Cartesian (x,y,z):

$$\begin{aligned} \Phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] \\ + \left[\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right)^2 \right] \\ - \frac{2}{3} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2 \end{aligned} \quad (1.45a)$$

Cylindrical (r, θ, z):

$$\begin{aligned} \Phi = 2 \left[\left(\frac{\partial v_r}{\partial r} \right)^2 + \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)^2 + \left(\frac{\partial v_z}{\partial z} \right)^2 \right. \\ \left. + \frac{1}{2} \left(\frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r} + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right)^2 + \frac{1}{2} \left(\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right)^2 \right. \\ \left. + \frac{1}{2} \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right)^2 - \frac{1}{3} (\nabla \cdot \mathbf{v})^2 \right] \end{aligned} \quad (1.45b)$$

Spherical (r, ϕ, θ):

$$\begin{aligned} \Phi = 2 \left\{ \left[\left(\frac{\partial v_r}{\partial r} \right)^2 + \left(\frac{1}{r} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} \right)^2 + \left(\frac{1}{r \sin \phi} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} + \frac{v_\phi \cot \phi}{r} \right)^2 \right] \right. \\ \left. + \frac{1}{2} \left[r \frac{\partial}{\partial r} \left(\frac{v_\phi}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \phi} \right]^2 + \frac{1}{2} \left[\frac{\sin \phi}{r} \frac{\partial}{\partial \phi} \left(\frac{v_\theta}{r \sin \phi} \right) + \frac{1}{r \sin \phi} \frac{\partial v_\theta}{\partial \theta} \right]^2 \right. \\ \left. + \frac{1}{2} \left[\frac{1}{r \sin \phi} \frac{\partial v_r}{\partial \theta} + r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2 \right\} - \frac{2}{3} (\nabla \cdot \mathbf{v})^2 \end{aligned} \quad (1.45c)$$

If the density does not vary significantly through the flow field, $\nabla \cdot \mathbf{v} = 0$ [eq. (1.6)] and the last term in each of expressions (1.45) vanishes.

It is worth reviewing the constant- ρ approximation that led to eq. (1.8) and recognizing that it differs conceptually from the “incompressible substance model” of thermodynamics. The latter is considerably more restrictive than the “nearly constant” density model, eq. (1.8). For example, a compressible substance such as air can flow in such a way that eq. (1.8) is a very good approximation of eq. (1.6).

For the restrictive class of fluids that are “incompressible” from the thermodynamic point of view, the specific heat at constant pressure c_p can be replaced by the lone specific heat of the fluid, c , on the left side of eq. (1.39). Water, liquid mercury, and engine oil are examples of fluids for which this substitution is justified. There are even convection problems in which the moving materials are actually solid (e.g., a roller and its substrate, in the zone of elastic contact). In such cases the $c_p = c$ substitution is permissible also.

It is important to note that the specific heat at constant volume c_v does not belong on the left side of eq. (1.39). This observation is important because Fourier [5,6], and later Poisson [7], who were the first to derive the energy equation for a convective flow, wrote c on the left side of eq. (1.39). They made this choice because their analyses were aimed specifically at *incompressible fluids* (liquids), for which c happens to have nearly the same value as c_p . Because of this choice, they did not have to account for the $P dV$ type of work done by the fluid packet as it expands or contracts in the flow field. In the modern era, however, the use of c_v instead of c_p is an error.

The prethermodynamics (caloric conservation) origins of the science of convection heat transfer are also responsible for the “thermal energy equation” label that some prefer to attach to eq. (1.39) without the $\beta T DP/Dt$ term. This terminology is sometimes used to stress (incorrectly) the conservation of “thermal” energy as something distinct from “mechanical and thermal” energy. In classical thermodynamics, however, this distinction disappeared as soon as the first law of thermodynamics was enunciated, that is, as soon as the thermodynamic property “energy” was defined, which happened in the years 1850–1851 (see Ref. 3, pp. 30–32).

Equation (1.39) represents the first law of thermodynamics. This law proclaims the conservation of the sum of energy change (the property) and energy interactions (heat transfer *and* work transfer). The suggestion that mechanical effects (e.g., work transfer) are absent from eq. (1.39) when the $\beta T DP/Dt$ term is absent is erroneous. The presence of c_p on the left side of the equation is the sign that each fluid packet expands or contracts (i.e., it does $P dV$ -type work) as it rides on the flow. The terms q''' and $\mu\Phi$ are work transfer rate terms also.

1.4 SECOND LAW OF THERMODYNAMICS

Any discussion of the basic principles of convective heat transfer must include the second law of thermodynamics, not because the second law is necessary for determining the flow and temperature field (it is not, because it is not an equation), but because the second law is the basis for much of the engineering *motive* (objective, purpose) for formulating and solving convection problems. For example, in the development of knowhow for the heat exchanger industry, we strive for improved thermal contact (enhanced heat transfer) *and* reduced pump power loss in order to improve the *thermodynamic* efficiency of the heat exchanger. Good heat exchanger design means, ultimately, efficient thermodynamic performance, that is, *minimum generation of entropy* or minimum destruction of exergy in the power/refrigeration system incorporating the heat exchanger [8–10]. For this reason, it is useful to review the second law and in this way to explain the commonsense origin of the engineering questions that led to today’s field of convective heat transfer.

The second law of thermodynamics states that all real-life processes are irreversible: In the case of a control volume, as in Fig. 1.1, this statement is

$$\frac{\partial S_{cv}}{\partial t} \geq \sum \frac{q_i}{T_i} + \sum_{\text{inlet ports}} \dot{m}s - \sum_{\text{outlet ports}} \dot{m}s \quad (1.46)$$

where S_{cv} is the instantaneous entropy inventory of the control volume, $\dot{m}s$ represents the entropy flows (streams) into and out of the control volume, and T_i is the absolute temperature of the boundary crossed by the heat transfer